

Video Transcript

Video 7: Querying a Database (04:40)

So, we're going to need to ask questions. So, the first thing that we need to do is to be able to get a grade. Let's just check what we get for database. So, this is our database now. And now, let's see if we can get the grade of a student. So, we're going to access this grades. So, 'def getGrade'. We need to pass in the database. We need to pass in the student ID and the problem ID. And now, we want to return what? Well, let's figure out what we're going to return. We need to access first from the database the grades. So, 'grades'. So, let's call this a row. So, from the row, let's suppose for 'row in db grades'. So, now we're picking out one of these. We're looping over the list and we're picking out a row. And from that row, what do we want?

Well, we want to return the points. So, we're going to need to get points and let's do a list comprehension. So, I need this is all going to be a list and we're going to have row points for row in database 'if'. Let's put that on the next line 'if'. Well, now we need row and we need the student ID. So, that's student is equal to 'sid' and row 'pset' is equal to pset ID, 'pid'. That looks good. Now we want to return this. Let's test it out. So, we're going to call getGrade and we'll pass in the database. And let's get student one, grade one. Let's just check it out. Now we got 98. And so, we're getting a row. Have we got two 98s? No. Yes, we have. Yes, we have. So, it got it right. We had a duplicate entry. Let's clean this up. And we need to re-zero our database. That's better, no getGrade. We have one grade 98.

So, be careful about re-running some of these cells. They'll just add duplicate data into the database. So, now we've got getGrade. I want you to change this. And instead of using a list comprehension, I want you to use a dictionary comprehension so that the result would look like this. So, I want this outputted instead of just the 98. I'd like this output as a dictionary. So, you need to change getGrade to have a dictionary comprehension. So, that's your first challenge.